

OpenMP Exam Guide

Basic Parallel

Question: Print Hello

```
#pragma omp parallel
{
    printf("Hello");
}
```

Simple Explanation: Creates multiple threads. Every thread runs the whole block.

Thread ID

Question: Print IDs

```
int id;
#pragma omp parallel private(id)
{
    id=omp_get_thread_num();
    printf("%d",id);
}
```

Simple Explanation: private gives each thread its own id. omp_get_thread_num returns thread number.

Parallel For

Question: Loop

```
#pragma omp parallel for
for(int i=0;i<10;i++) printf("%d",i);
```

Simple Explanation: Loop iterations are divided among threads.

Sum

Question: Find sum

```
int sum=0;
#pragma omp parallel for reduction(+:sum)
for(int i=0;i<n;i++) sum+=arr[i];
```

Simple Explanation: Each thread has a private sum. OpenMP adds them together.

Factorial

Question: Find factorial

```
long fact=1;
#pragma omp parallel for reduction(*:fact)
for(int i=1;i<=n;i++) fact*=i;
```

Simple Explanation: Same skeleton as sum; only + changes to *.

Maximum

Question: Find max

```
int best=arr[0];
#pragma omp parallel for reduction(max:best)
for(int i=1;i<n;i++) if(arr[i]>best) best=arr[i];
```

Simple Explanation: Each thread finds local max, OpenMP combines.

Minimum

Question: Find min

```
int best=arr[0];
#pragma omp parallel for reduction(min:best)
```

```
for(int i=1;i<n;i++) if(arr[i]<best) best=arr[i];
```

Simple Explanation: Each thread finds local min, OpenMP combines.